THE INTERNAL HOM FROM A K-FLAT COMPLEX INTO A K-INJECTIVE COMPLEX IS AGAIN K-INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the displayed four-term chain of equalities with the reason for each, and the final vanishing from K-flatness of the tensor factor and K-injectivity of the target.

Editorial note. The label lemma-RHom-from-K-flat-into-K-injective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat and K-injective complexes of modules on a ringed site. Same chapter, the definitions of K-flat and K-injective complexes used as the two hypotheses of the statement. Supplies the properties invoked at the end of the proof: the total tensor product with a K-flat complex of an acyclic complex is acyclic, and a K-injective complex has no homotopy classes of maps from acyclic complexes.
- The global cohomology description of Hom complexes. Same chapter, the equation identifying Hom in the homotopy category with H^0 of the global sections of the internal hom complex. Supplies the first and third equalities of the displayed chain.
- Composition of internal homs and the total tensor product. Same chapter, the lemma identifying the internal hom into an internal hom with the internal hom from the total tensor product. Supplies the second equality of the displayed chain.

Source reference. Original reference 06XZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2999–2999. Prerequisite pointer invoked at proof line(s) 8824; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original equation 0A8Z: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8608–8608. Prerequisite pointer invoked at proof line(s) 8818; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A5Y: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8616–8616. Prerequisite pointer invoked at proof line(s) 8819; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{I}^\bullet$ be a K-injective complex of $\mathcal{O}$-modules. Let $\mathcal{L}^\bullet$ be a K-flat complex of $\mathcal{O}$-modules. Then $\mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)$ is a K-injective complex of $\mathcal{O}$-modules.*

Proof. Namely, if $\mathcal{K}^\bullet$ is an acyclic complex of $\mathcal{O}$-modules, then

$$\begin{aligned}
\mathrm{Hom}_{K(\mathcal{O})}(\mathcal{K}^\bullet, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)) &= H^0(\Gamma(\mathcal{C}, \mathcal{H}om^\bullet(\mathcal{K}^\bullet, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)))) \\
&= H^0(\Gamma(\mathcal{C}, \mathcal{H}om^\bullet(\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet), \mathcal{I}^\bullet))) \\
&= \mathrm{Hom}_{K(\mathcal{O})}(\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet), \mathcal{I}^\bullet) \\
&= 0
\end{aligned}$$

The first equality by (0A8Z). The second equality by Lemma 0A5Y. The third equality by (0A8Z). The final equality because $\mathrm{Tot}(\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{L}^\bullet)$ is acyclic because $\mathcal{L}^\bullet$ is K-flat (Definition 06XZ) and because $\mathcal{I}^\bullet$ is K-injective. □